

MTH-100

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Question No 1

Let $F(x) = \frac{2x^2 - 5x}{x^2 - 3}$ and $G(x) = \sqrt{19 - x^2}$

a) Determine The domain and range of each functions :

Domain:

To find the domain of $f(x)$ we need to ensure that the denominator $x^2 - 3$ is not zero, as division by zero is undefined.

1. Set the denominator equal to zero:

$$x^2 - 3$$

$$x^2 = 3$$

$$x = \pm\sqrt{3}$$

2. So, $f(x)$ is undefined at $x = \sqrt{3}$ and $x = -\sqrt{3}$

• **Domain:**

$$(-\infty, -\sqrt{3}) \cup (-\sqrt{3}, \sqrt{3}) \cup (\sqrt{3}, \infty)$$

Range:

Since this is a rational function with a cubic polynomial in the numerator and a quadratic polynomial in the denominator, it will approach positive or negative infinity as $x \rightarrow -\infty$ and $x \rightarrow \infty$. However, determining the exact range analytically is complex and typically requires calculus (finding asymptotes, critical points, etc.). Therefore, we can say:

- Range: $(-\infty, \infty)$ except possibly at values excluded by vertical asymptotes or points of discontinuity.

2. Function : $G(x) = \sqrt{19 - x^2}$

Domain:

For $g(x)$ to be defined, the expression inside the square root must be non-negative:

1. Set the inside of the square root to be greater than or equal to zero:

$$19 - x^2 \geq 0$$

$$x^2 \leq 19$$

$$-\sqrt{19} \leq x \leq \sqrt{19}$$

2. So, $g(x)$ is defined for x in the interval $[-\sqrt{19}, \sqrt{19}]$

- Domain: $[-\sqrt{19}, \sqrt{19}]$

Range:

Since $g(x) = \sqrt{19 - x^2}$ represents the upper half of a circle with radius $\sqrt{19}$, the values of $g(x)$ range from 0 to $\sqrt{19}$

Range: $(0, \sqrt{19})$

Summary

- For $f(x) = \frac{2x^3 - 5x}{x^2 - 3}$
- Domain: $(-\infty, -\sqrt{3}) \cup (-\sqrt{3}, \sqrt{3}) \cup (\sqrt{3}, \infty)$
- Range: $(-\infty, \infty)$ (except possibly at points of discontinuity)
- For $g(x) = \sqrt{19 - x^2}$
- Domain: $(-\sqrt{19}, \sqrt{19})$
- Range: $(0, \sqrt{19})$

b) Classify each function as even odd or neither justify your answer with the necessary algebraic Checks

To determine whether each function is even, odd, or neither, we need to apply the definitions of even and odd functions.

- A function $f(x)$ is even if $f(-x) = f(x)$ for all x in its domain.
- A function $f(x)$ is odd if $f(-x) = -f(x)$ for all x in its domain.
- If neither of these conditions is satisfied, the function is classified as neither even nor odd.

Let's check each function.

1. Function $F(x)$

To determine if $f(x)$ is even, odd, or neither, let's calculate $f(-x)$

1. Substitute $-x$ for x in $f(x)$:

$$F(-x) = \frac{2(-x^3) - 5(-x)}{(-x)^2 - 3}$$

2. Simplify $f(-x)$

$$F(-x) = \frac{2(-x^3) + 5x}{x^2 - 3}$$

$$= \frac{-2x^3 + 5x}{x^2 - 3}$$

3. Compare $f(x)$ to $f(-x)$:

- $f(x) \neq f(-x)$
- $f(x) \neq -f(-x)$

Since $f(-x) \neq f(x)$ and $f(-x) \neq -f(x)$, the function $f(x)$ is neither even nor odd.

2. Function $g(x)$

To determine if $g(x)$ is even, odd, or neither, let's calculate $g(-x)$.

1. Substitute $-x$ for x in $g(x)$:

$$G(-x) = \sqrt{19 - (-x)^2}$$

2. Simplify $g(-x)$

$$G(-x) = \sqrt{19 - x^2}$$

3. Compare $g(-x)$ to $g(x)$:

- $g(x) = \sqrt{19 - x^2}$
- $g(-x) = \sqrt{19 - x^2}$

Since $g(-x) = g(x)$ for all x in the domain, $g(x)$ is even.

Summary

- For $f(x)$: The function is neither even nor odd.
- For $g(x)$: The function is even.

C) For $f(g(x))$ and $g(f(z))$ Find the domains and explain any restrictions

To find the domains of $f(g(x))$ and $g(f(z))$, we need to understand the compositions of the functions $f(x)$ and $g(x)$, and determine any restrictions that arise in these compositions.

1. Finding the Domain of $f(g(x))$

To form $f(g(x))$, we need to substitute $g(x)$ into $f(x)$. The domain of $f(g(x))$ will depend on both:

- The domain of $g(x)$, since $g(x)$ must be defined.

- The domain of $f(x)$ evaluated at $g(x)$, since $f(g(x))$ must be defined.

Step 1: Domain of $g(x) = \sqrt{19 - x^2}$

As previously calculated:

- The domain of $g(x)$ is $x \in (-\sqrt{19}, \sqrt{19})$ since $19 - x^2 \geq 0$

Step 2: Substitute $g(x)$ into $f(x)$ to form $f(g(x))$

Since $f(x) = \frac{2x^3 - 5x}{x^2 - 3}$ substituting $G(x) = \sqrt{19 - x^2}$ into $f(x)$ gives:

$$F(g(x)) = f(\sqrt{19 - x^2}) = \frac{2(\sqrt{19 - x^2})^3 - 5(\sqrt{19 - x^2})}{(\sqrt{19 - x^2})^2 - 3}$$

Step 3: Determine Additional Restrictions from $\frac{f(g(x))}{f(g(x))}$

To find additional restrictions, we need to ensure that the denominator in $f(g(x))$ does not equal zero:

$$(\sqrt{19 - x^2})^2 - 3 \neq 0$$

$$19 - x^2 - 3 \neq 0$$

$$16 - x^2 \neq 0$$

$$x^2 \neq 16$$

$$x \neq \pm 4$$

Thus, $x = \pm 4$ are values that would make the denominator zero, so they are excluded from the domain of $f(g(x))$

Domain of $f(g(x))$

Combining the restrictions from the domain of $g(x)$ and the restriction $x \neq \pm 4$, we get:

- Domain of $f(g(x))$

$$x \in (-\sqrt{19}, \sqrt{19}) \setminus (-4, 4)$$

2. Finding the Domain of $g(f(z))$

To form $g(f(z))$ we need to substitute $f(z)$ into $g(x)$. The domain of $g(f(z))$ will depend on both:

- The domain of $f(z)$, since $f(z)$ must be defined.
- The domain of $g(x)$ evaluated at $f(z)$, since $g(f(z))$ must be defined.

Step 1: Domain of $F(z) = \frac{2z^3 - 5z}{z^2 - 3}$

As previously calculated:

- The domain of $f(z)$ is $z \in (-\infty, -\sqrt{3}) \cup (-\sqrt{3}, \sqrt{3}) \cup (\sqrt{3}, \infty)$, since $z = \pm\sqrt{3}$ would make the denominator zero.

Step 2: Substitute $f(z)$ into $g(x)$ to form $g(f(z))$

Since $g(x) = \sqrt{19 - x^2}$, substituting $f(z)$ into $g(x)$ gives:

$$G(f(z)) = g\left(\frac{2z^3 - 5z}{z^2 - 3}\right) = \sqrt{19 - \left(\frac{2z^3 - 5z}{z^2 - 3}\right)^2}$$

Step 3: Determine Additional Restrictions from $g(f(z))$

To ensure $g(f(z))$ is defined, the expression inside the square root must be non-negative:

$$19 - \left(\frac{2z^3 - 5z}{z^2 - 3}\right)^2 \geq 0$$

This inequality is complex to solve algebraically, but a key observation is that we need:

$$\left| \frac{2z^3 - 5z}{z^2 - 3} \right| \leq \sqrt{19}$$

Furthermore, $f(z)$ must be in the domain of $g(x)$ which is $(-\sqrt{19}, \sqrt{19})$. This adds a restriction to the values of z , limiting $f(z)$ within that range. A precise determination of this subset of z -values would typically require numerical or graphical analysis.

Domain of $g(f(z))$

In general terms, the domain of $g(f(z))$ includes values of z in the domain of $f(z)$ such that $f(z)$ is in the domain of $g(x)$ and excludes points where the expression inside the square root becomes negative or where $f(z)$ itself is undefined.

To solve the system of equations using the Reduced Row Echelon Form (RREF) method, let's start by writing the system as an augmented matrix:

The given system is:

$$2x + 3y = 5$$

$$4x + y + 2z = 6$$

$$-x + 5y + 3z = -4$$

We can rewrite this system in augmented matrix form:

$$\left(\begin{array}{ccc|c} 2 & 3 & 0 & 5 \\ 4 & 1 & 2 & 6 \\ -1 & 5 & 3 & -4 \end{array} \right)$$

Now, we will use row operations to transform this matrix into Reduced Row Echelon Form (RREF).

Step 1: Make the entry in the first row, first column a 1

To get a 1 in the first row, first column, we can divide the entire first row by 2:

$$\left(\begin{array}{ccc|c} 1 & \frac{3}{2} & 0 & \frac{5}{2} \\ 4 & 1 & 2 & 6 \\ -1 & 5 & 3 & -4 \end{array} \right)$$

Step 2: Make the entries below the pivot in the first column zero

To make the entry in the second row, first column zero, subtract 4 times the first row from the second row. For the third row, add the first row.

1. Row 2: $R_2 - 4 \times R_1$
2. Row 3: $R_3 + R_1$

$$\left(\begin{array}{ccc|c} 1 & \frac{3}{2} & 0 & \frac{5}{2} \\ 0 & -5 & 2 & -4 \\ 0 & \frac{13}{2} & 3 & -\frac{3}{2} \end{array} \right)$$

Step 3: Make the entry in the second row, second column a 1

To make the entry in the second row, second column a 1, divide the entire second row by -5:

$$\left(\begin{array}{ccc|c} 1 & \frac{3}{2} & 0 & \frac{5}{2} \\ 0 & 1 & -\frac{2}{5} & \frac{4}{5} \\ 0 & \frac{13}{2} & 3 & -\frac{3}{2} \end{array} \right)$$

Step 4: Make the entries above and below the pivot in the second column zero

1. To make the entry in the first row, second column zero, subtract $\frac{3}{2}$ times the second row from the first row.

2. To make the entry in the third row, second column zero, subtract $\frac{13}{2}$ times the second row from the third row.

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{3}{5} & \frac{7}{5} \\ 0 & 1 & -\frac{2}{5} & \frac{4}{5} \\ 0 & 0 & \frac{37}{5} & -\frac{13}{5} \end{array} \right)$$

Step 5: Make the entry in the third row, third column a 1

To make the entry in the third row, third column a 1, divide the entire third row by $\frac{37}{5}$:

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{3}{5} & \frac{7}{5} \\ 0 & 1 & \frac{2}{5} & \frac{4}{5} \\ 0 & 0 & 1 & -\frac{13}{37} \end{array} \right)$$

Step 6: Make the entries above the pivot in the third column zero

1. To make the entry in the first row, third column zero, subtract $\frac{3}{5}$ times the third row from the first row.
2. To make the entry in the second row, third column zero, add $\frac{2}{5}$ times the third row to the second row.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{10}{37} \\ 0 & 1 & 0 & \frac{29}{37} \\ 0 & 0 & 1 & -\frac{13}{37} \end{array} \right)$$

Solution

From the final matrix, we can read off the solutions for x,y,z

$$x = \frac{10}{37}, y = \frac{29}{37}, z = -\frac{13}{37}$$